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Time on Task:

5 hours, 18 minutes

Progress:

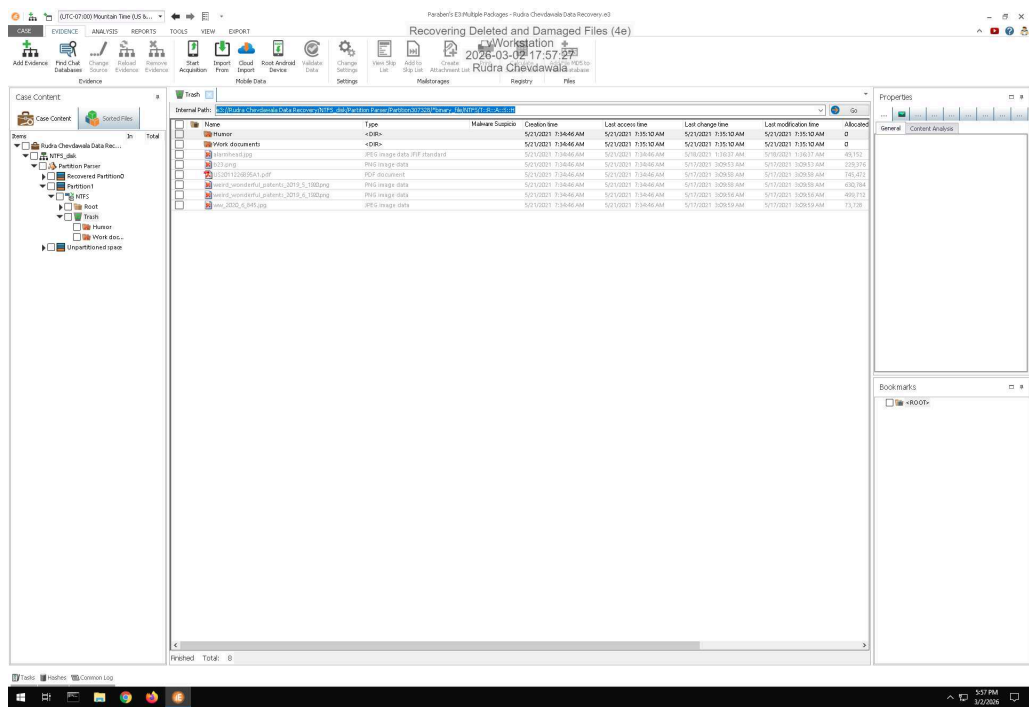
100%

Report Generated: Tuesday, March 3, 2026 at 10:32 PM

Section 1: Hands-On Demonstration

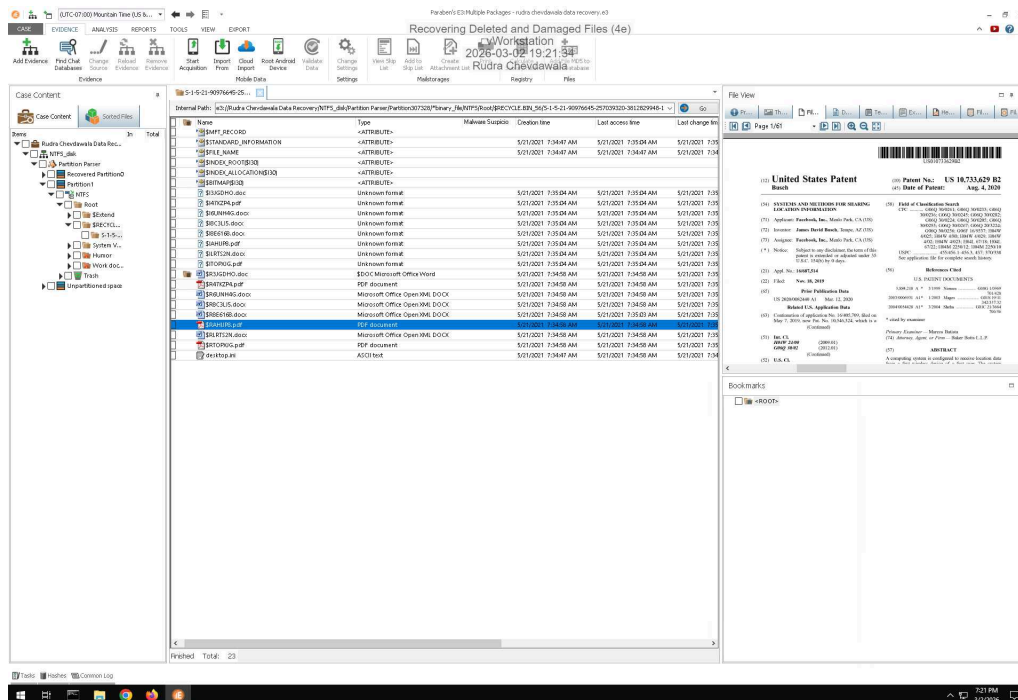
Part 1: Recover Deleted Files from an NTFS Drive Image with E3

13. Make a screen capture showing the list of recovered files and folders in the E3 Trash folder.

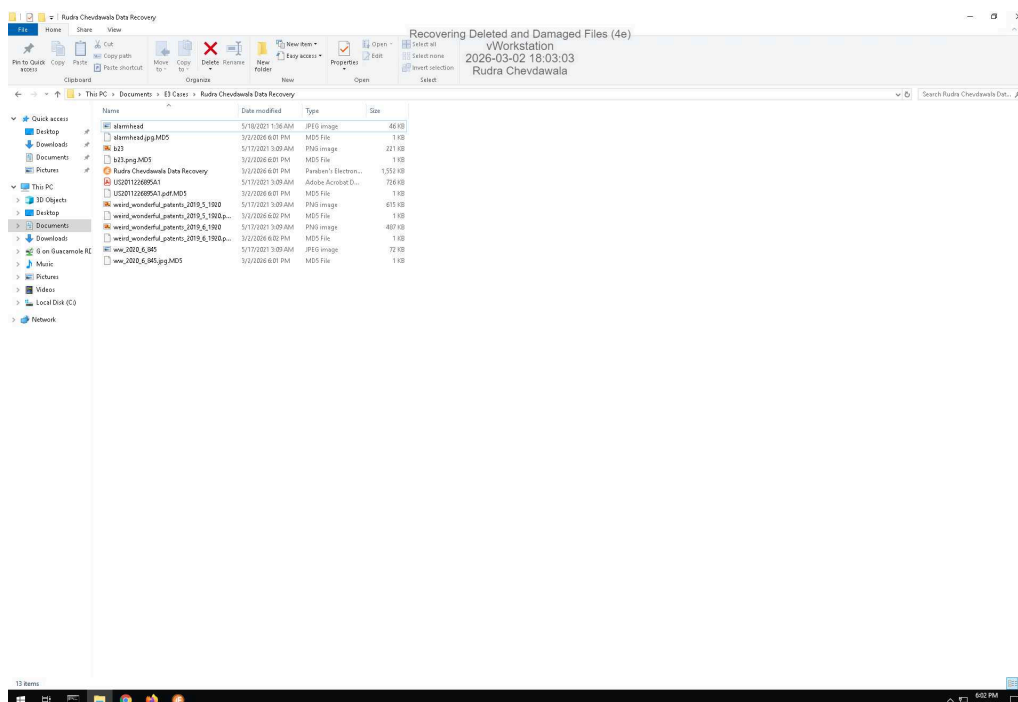


## Digital Forensics, Investigation, and Response, Fourth Edition - Lab 03

20. **Make a screen capture** showing the patent file in the File Viewer.



25. **Make a screen capture** showing the **recovered files** in the **File Explorer**.

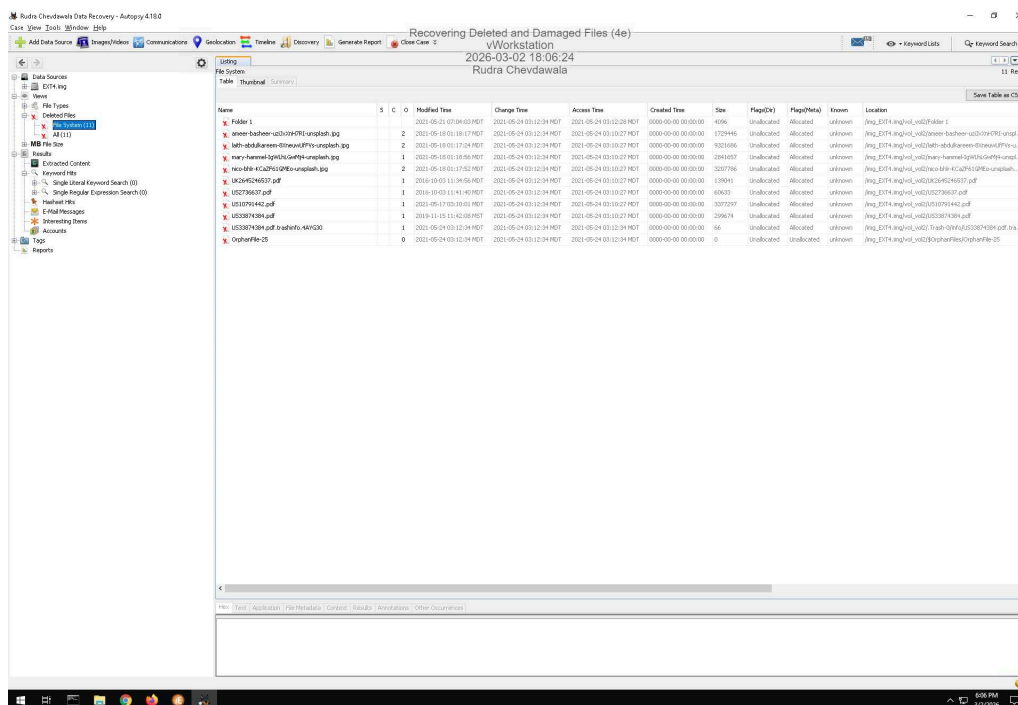


## Part 2: Recover Deleted Files from an Ext4 Drive Image with Autopsy

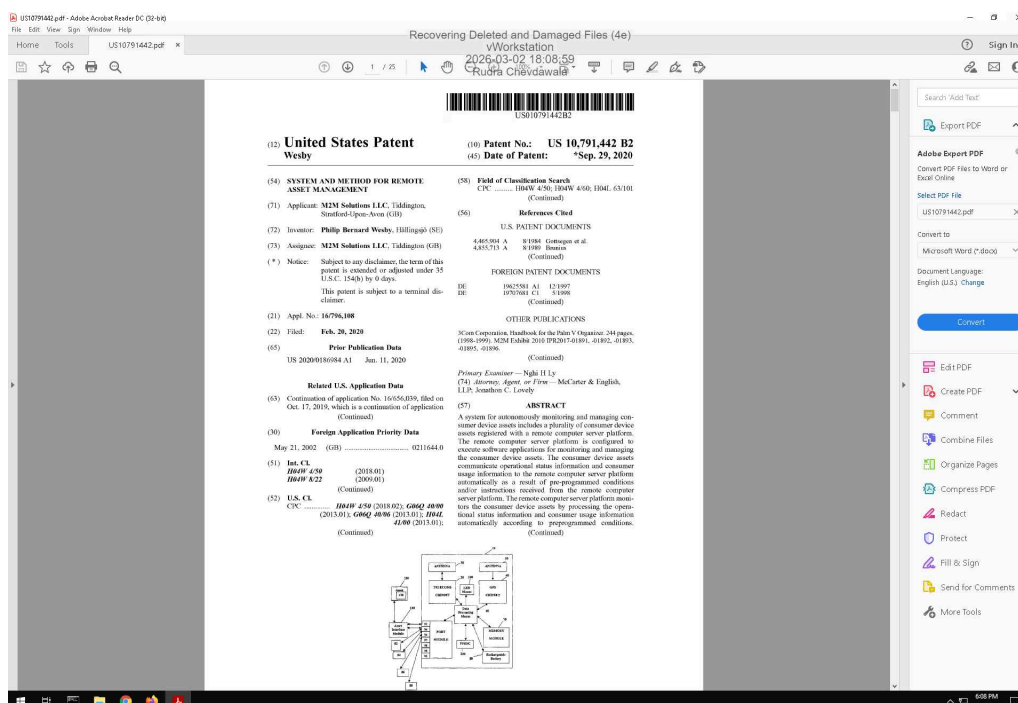
# Recovering Deleted and Damaged Files (4e)

## Digital Forensics, Investigation, and Response, Fourth Edition - Lab 03

### 14. Make a screen capture showing the contents of the list of deleted files in Autopsy.



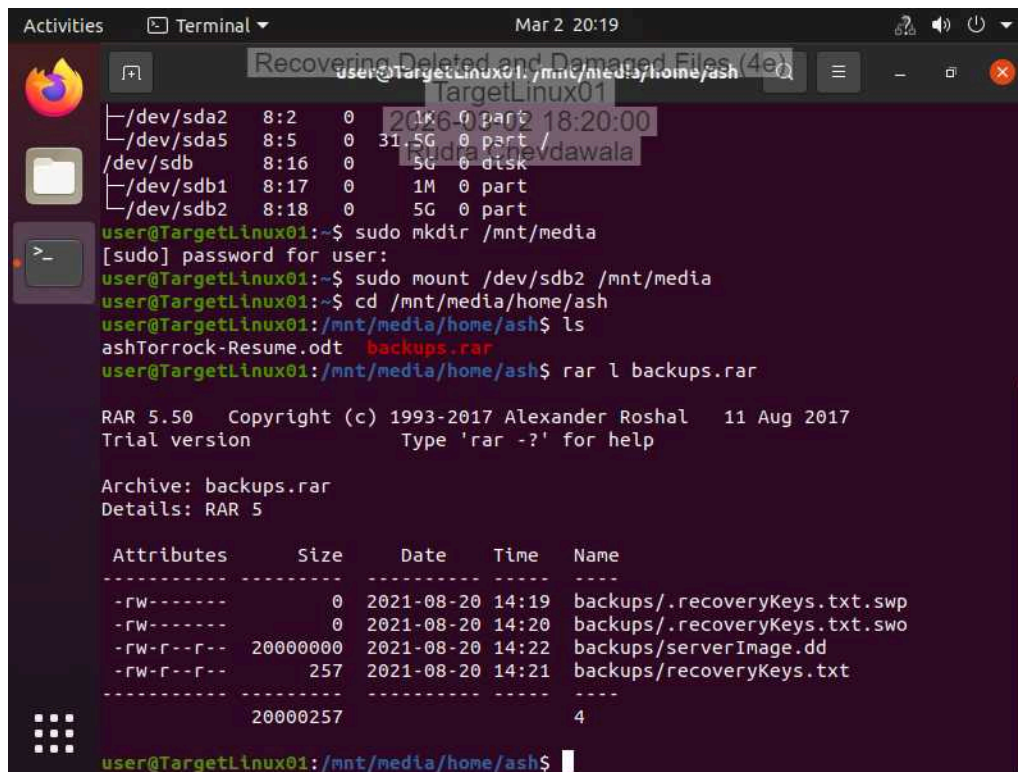
### 22. Make a screen capture showing the recovered patent file.



## Section 2: Applied Learning

### Part 1: Recover Deleted Files in Linux with PhotoRec

9. Make a screen capture showing the contents of the RAR archive in the /mnt/media/home/ash directory.



```
user@TargetLinux01: /mnt/media/home/ash
/dev/sda2  8:2  0  1M  0 part
/dev/sda5  8:5  0  31.5G  0 part /
/dev/sdb   8:16 0  5G  0 disk
/dev/sdb1  8:17 0  1M  0 part
/dev/sdb2  8:18 0  5G  0 part
user@TargetLinux01:~$ sudo mkdir /mnt/media
[sudo] password for user:
user@TargetLinux01:~$ sudo mount /dev/sdb2 /mnt/media
user@TargetLinux01:~$ cd /mnt/media/home/ash
user@TargetLinux01:/mnt/media/home/ash$ ls
ashTorrock-Resume.odt  backups.rar
user@TargetLinux01:/mnt/media/home/ash$ rar l backups.rar

RAR 5.50 Copyright (c) 1993-2017 Alexander Roshal 11 Aug 2017
Trial version Type 'rar -?' for help

Archive: backups.rar
Details: RAR 5

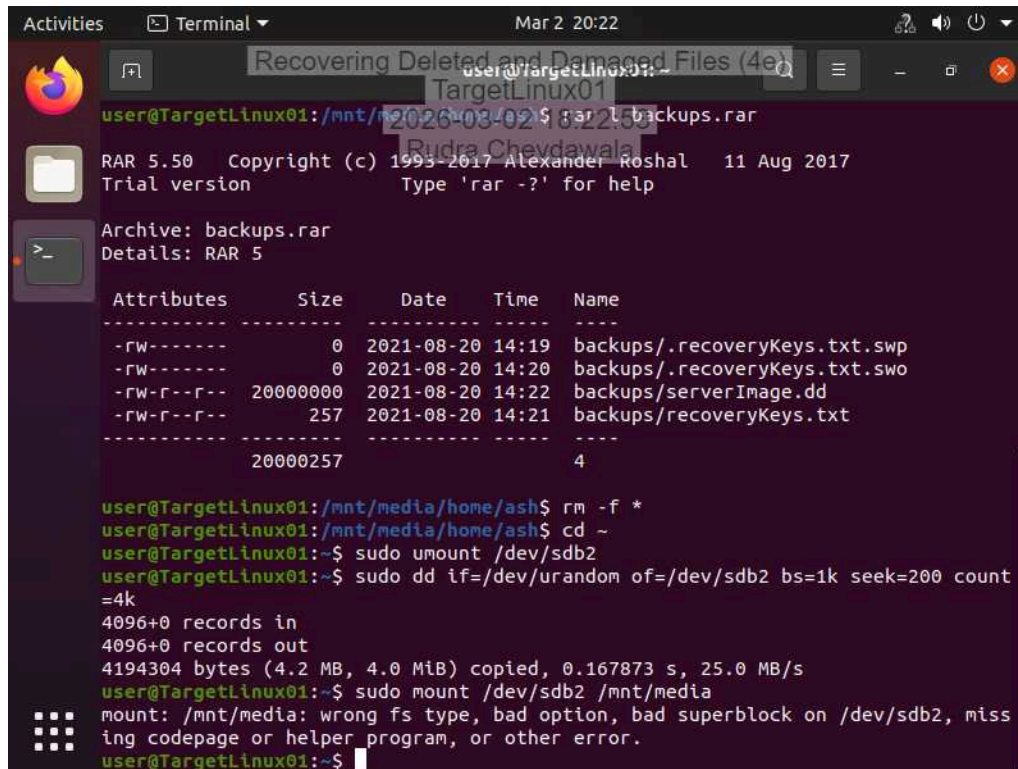
Attributes      Size      Date      Time      Name
-----
-rw-----      0      2021-08-20 14:19 backups/.recoveryKeys.txt.swp
-rw-----      0      2021-08-20 14:20 backups/.recoveryKeys.txt.swo
-rw-r--r-- 20000000 2021-08-20 14:22 backups/serverImage.dd
-rw-r--r--    257 2021-08-20 14:21 backups/recoveryKeys.txt
-----
                20000257                4

user@TargetLinux01:/mnt/media/home/ash$
```

## Recovering Deleted and Damaged Files (4e)

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15. Make a screen capture showing the failed mount attempt on the /dev/sdb2 device.



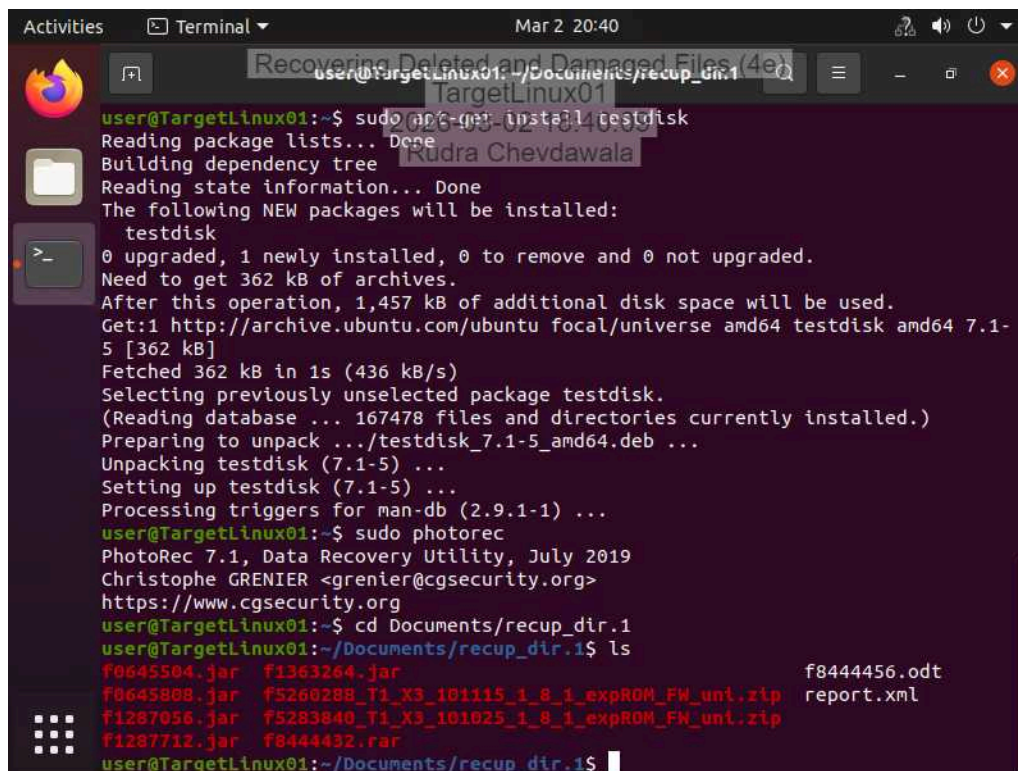
```
user@TargetLinux01:~$ rar x backups.rar
RAR 5.50 Copyright (c) 1993-2017 Alexander Roshal 11 Aug 2017
Trial version
Type 'rar -?' for help

Archive: backups.rar
Details: RAR 5

Attributes      Size      Date      Time      Name
-----
-rw-----      0      2021-08-20 14:19 backups/.recoveryKeys.txt.swp
-rw-----      0      2021-08-20 14:20 backups/.recoveryKeys.txt.swo
-rw-r--r-- 20000000 2021-08-20 14:22 backups/serverImage.dd
-rw-r--r--    257      2021-08-20 14:21 backups/recoveryKeys.txt
-----
                20000257          4

user@TargetLinux01:/mnt/media/home/ash$ rm -f *
user@TargetLinux01:/mnt/media/home/ash$ cd ~
user@TargetLinux01:~$ sudo umount /dev/sdb2
user@TargetLinux01:~$ sudo dd if=/dev/urandom of=/dev/sdb2 bs=1k seek=200 count=4k
4096+0 records in
4096+0 records out
4194304 bytes (4.2 MB, 4.0 MiB) copied, 0.167873 s, 25.0 MB/s
user@TargetLinux01:~$ sudo mount /dev/sdb2 /mnt/media
mount: /mnt/media: wrong fs type, bad option, bad superblock on /dev/sdb2, missing codepage or helper program, or other error.
user@TargetLinux01:~$
```

32. Make a screen capture showing the compressed files recovered by PhotoRec.



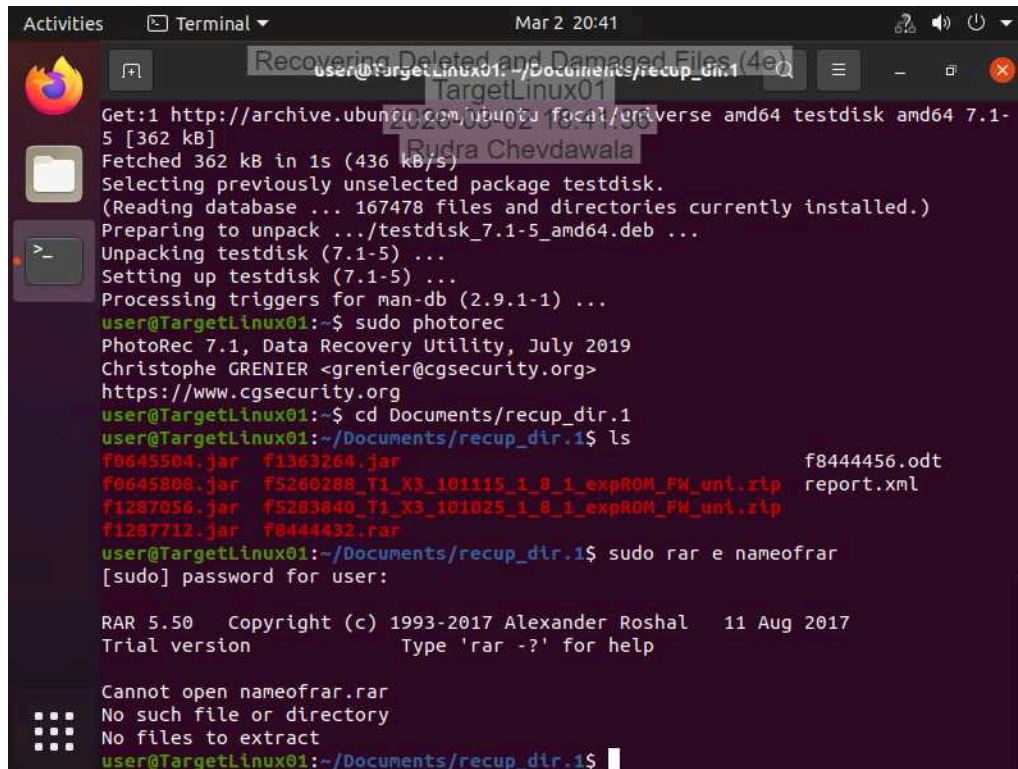
```
user@TargetLinux01:~$ sudo apt-get install testdisk
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  testdisk
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 362 kB of archives.
After this operation, 1,457 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu focal/universe amd64 testdisk amd64 7.1-5 [362 kB]
Fetched 362 kB in 1s (436 kB/s)
Selecting previously unselected package testdisk.
(Reading database ... 167478 files and directories currently installed.)
Preparing to unpack .../testdisk_7.1-5_amd64.deb ...
Unpacking testdisk (7.1-5) ...
Setting up testdisk (7.1-5) ...
Processing triggers for man-db (2.9.1-1) ...
user@TargetLinux01:~$ sudo photorec
PhotoRec 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
user@TargetLinux01:~$ cd Documents/recup_dir.1
user@TargetLinux01:~/Documents/recup_dir.1$ ls
f0645504.jar  f1363264.jar  f8444456.odt
f0645808.jar  f5260288_T1_X3_101115_1_8_1_expROM_FW_unl.zip  report.xml
f1287056.jar  f5283840_T1_X3_101025_1_8_1_expROM_FW_unl.zip
f1287712.jar  f0444432.rar
user@TargetLinux01:~/Documents/recup_dir.1$
```



## Recovering Deleted and Damaged Files (4e)

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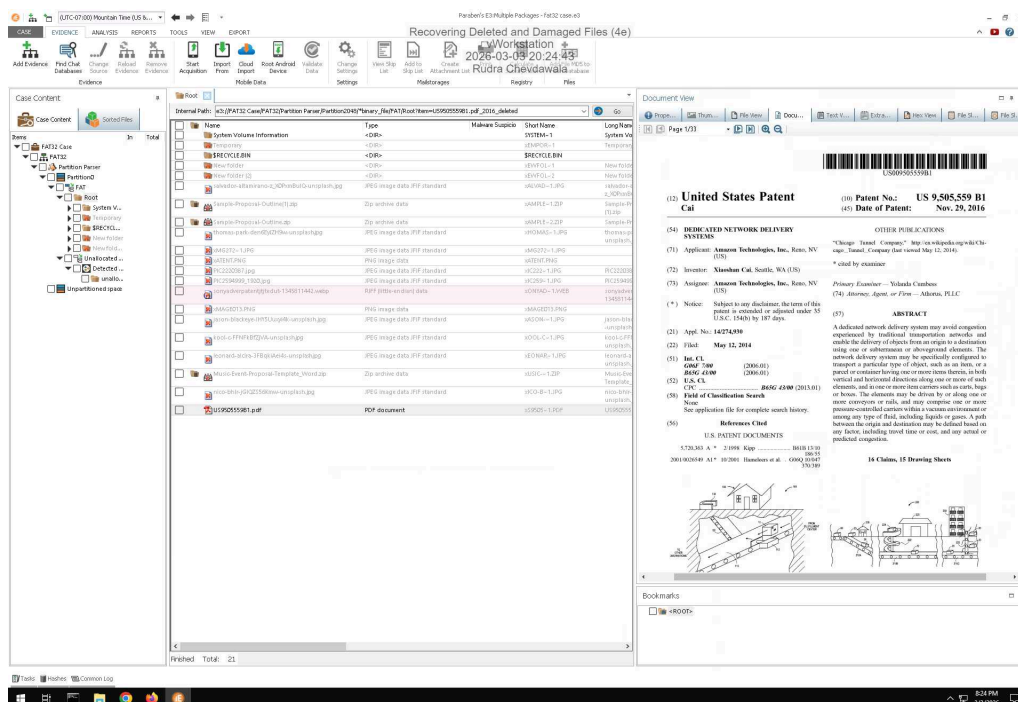
35. Make a screen capture showing the backup files recovered from the RAR archive.

A terminal window titled 'Terminal' with a date and time of 'Mar 2 20:41'. The user is logged in as 'user' on a machine named 'TargetLinux01'. The terminal shows the following commands and output:  
1. `Get:1 http://archive.ubuntu.com/ubuntu focal/universe amd64 testdisk amd64 7.1-5 [362 kB]`  
2. `Fetched 362 kB in 1s (436 kB/s)`  
3. `Selecting previously unselected package testdisk.`  
4. `(Reading database ... 167478 files and directories currently installed.)`  
5. `Preparing to unpack .../testdisk_7.1-5_amd64.deb ...`  
6. `Unpacking testdisk (7.1-5) ...`  
7. `Setting up testdisk (7.1-5) ...`  
8. `Processing triggers for man-db (2.9.1-1) ...`  
9. `user@TargetLinux01:~$ sudo photorec`  
10. `PhotoRec 7.1, Data Recovery Utility, July 2019`  
11. `Christophe GRENIER <grenier@cgsecurity.org>`  
12. `https://www.cgsecurity.org`  
13. `user@TargetLinux01:~$ cd Documents/recup_dir.1`  
14. `user@TargetLinux01:~/Documents/recup_dir.1$ ls`  
15. Output of `ls`:  
f0645504.jar f1363264.jar f8444456.odt  
f0645808.jar f5260288\_T1\_X3\_101115\_1\_8\_1\_expROM\_FW\_unl.zip report.xml  
f1287056.jar f5283840\_T1\_X3\_101025\_1\_8\_1\_expROM\_FW\_unl.zip  
f1287712.jar f8444432.rar  
16. `user@TargetLinux01:~/Documents/recup_dir.1$ sudo rar e nameofrar`  
17. `[sudo] password for user:`  
18. `RAR 5.50 Copyright (c) 1993-2017 Alexander Roshal 11 Aug 2017`  
19. `Trial version Type 'rar -?' for help`  
20. `Cannot open nameofrar.rar`  
21. `No such file or directory`  
22. `No files to extract`  
23. `user@TargetLinux01:~/Documents/recup_dir.1$`

### Section 3: Challenge and Analysis

#### Part 1: Recover Deleted Files from a FAT Drive Image

Make a screen capture showing the patent file recovered from the FAT32 drive image within E3.



#### Part 2: Recover Deleted Files from a APFS Drive Image

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